THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

December 13, 2016 TIME: 2:30p.m.-5:00p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL EIGHT questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- If you do not otherwise specify, you are assumed to be answering the questions in order. In particular, indicate on each question where your answer to the next question is if it does not follow immediately. (For example you may write 'turn back 3 pages for Question 5' at the end of Question 4.)
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.

1. (20 marks) Give answers only to the following questions. Explanation is **NOT** required. Each question carries 4 marks.

   (a) Find the critical point of $f(x, y) = xy + 2x - \ln(x^2 y)$ in the open first quadrant $x > 0$ and $y > 0$. Also determine the nature of this critical point (relative maximum, relative minimum or saddle point).

   (b) Let $f(x, y) = -x^2 y + y^2 + 2y$. Using ONE iteration of Newton-Raphson's method, estimate the minimum of f with the initial guess $(-3, -3)$. (Round your answers to 2 decimal places.)

   (c) The specific gravity S of a solid may be calculated by the formula $S = \frac{x}{x-y}$, where x is the weight (in *Newtons*) of the solid in air and y is the weight (in *Newtons*) of the solid in water. Suppose that in a certain experiment, x is measured to be 9 with an error not exceeding 0.05 and y is measured to be 5 with an error not exceeding 0.1. If ΔS denotes the error in calculating S by these measurements, use total differential to estimate the maximum of ΔS.(Round your answers to 2 decimal places.)

   (d) Let $w = x^2 e^{2y} \cos(3z)$. Find the value of dw/dt at the point $(1, \ln 2, 0)$ on the curve $x = \cos t$, $y = \ln(t + 2)$, $z = t$.

   (e) Evaluate the volume of D where D is the solid right cylinder whose base is the region between the circles $r = \cos\theta$ and $r = 2\cos\theta$ and whose top lies in the plane $z = 3 - y$.

2. (20 marks) In each multiple choice question below, some numbered choices are given and some are correct (wordings like is and are do not indicate the singularity /plurality of the number of correct choices). Answer the question by adding up the numbers of the correct choices. In case all choices are wrong, answer 0. Explanation is **NOT** required. Each question carries 2 marks.

EXAMPLE: Which of the following is correct?

(1) $1 + 2 = 3$.

(2) $1 + 1 = 2$.

(4) $1/0 = 1$.

(8) $1 - 1 = 0$. Answer: 11.

(a) Which of the following matrices is in row echelon form?

(1) $\begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \end{bmatrix}$.

(2) $\begin{bmatrix} 0 & 1 & 1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

(4) The 3×4 zero matrix.

(8) The 3×3 identity matrix.

(b) For a homogeneous system of 4 linear equations with 3 unknowns, which of the following is possible?

(1) The solution set is empty.

(2) The solution set contains unique solution.

(4) The solution set contains unique non-trivial solution.

(8) The solution set contains infinitely many solutions.

(c) Let A be a square matrix. Then A and A^T must have the same

(1) rank.

(2) determinant.

(4) set of eigenvalues.

(8) eigenvectors.

(d) Which of the following matrices is invertible?

(1) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$.

(2) $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 11 & 13 & 15 & 17 \\ 111 & 113 & 115 & 117 \\ 1111 & 1113 & 1115 & 1117 \end{bmatrix}$.

(4) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \\ 1 & 8 & 27 & 64 \\ 1 & 16 & 81 & 256 \end{bmatrix}$.

(8) $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$.

(e) If A and B are both upper triangular $n \times n$ matrices , then

(1) AB is also upper triangular.

(2) The diagonal entries of A are eigenvalues of A.

(4) $A^T B^T$ is lower triangular.

(8) $\det A$ is equal to the product of diagonal entries of A.

(f) Let A and B be square matrices of the same size. If A and B are both

_____ , so is AB.

(1) invertible.

(2) symmetric.

(4) upper triangular.

(8) singular.

(g) Which of the following is a linear combination of the vectors $(1, 2, 0)^T$ and $(0, 1, 2)^T$?

(1) $(1, 3, 2)^T$.

(2) $(-1, 0, 4)^T$.

(4) $(0, 1, 2)^T$.

(8) $(1, 0, 2)^T$.

(h) If A is a 4×6 matrix with rank 3, then the reduced row echelon form of A has

(1) at least one zero row.

(2) at least one zero column.

(4) exactly two zero columns.

(8) at most three zero columns.

(i) Which of the following operation will not change the determinant of a matrix?

(1) interchange two rows.

(2) multiply a row by a non-zero constant.

(4) add a multiple of a row to another row.

(8) add a multiple of a column to another column.

(j) If A and B are both $n \times n$ matrices with rank k, which of the following must be true?

(1) $\text{rank}(A + B) = k$.

(2) $\text{rank}(A + B) \leq k$.

(4) $\text{rank}(A + B) = \text{rank}(A - B)$.

(8) $\text{rank}(A + B) = \text{rank}(B + A)$.

3. (10 marks) It is known that the temperature T at any point (x, y, z) is given by $T = 2x^2 + 5y^2 + 11z^2 + 20xy - 4xz + 16yz$. Find the hottest point(s) on the unit sphere $x^2 + y^2 + z^2 = 1$.

4. (10 marks) Evaluate the integral

$$\int_0^2 \int_{y/2}^{(y+4)/2} (2x - y)y^3 e^{(2x-y)^2} dx dy.$$

(Hint: You can use the transformation: $x = u + v/2$ and $y = v$.)

5. (10 marks) Find the volume of the solid which is bounded below by $z = \sqrt{3x^2 + 3y^2}$ and bounded above by $x^2 + y^2 + z^2 = 16$.

6. (10 marks) Find the mass of a circular cone of base radius $r = 1$ and height $h = 2$ given that the density equals the distance from the vertex.

7. (10 marks) Solve the system of linear equations

$$\begin{cases} x_1 - 2x_2 + 3x_3 = 6 \\ 2x_1 - 3x_2 + 2x_3 = 2 \\ x_1 + 2x_2 - 5x_3 = -10 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

8. (10 marks) Given that

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

(a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix P such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate A^{2016}.

******* **END OF PAPER** *******